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So far, we have studied simple linear regression, i.e. regression with one predictor variable x.

$$y = \beta_0 + \beta_1 x + \varepsilon$$

In practice, we often have access to additional variables that we may choose to use in order to predict the response (y).

next: multiple predictors

Suppose we're interested in modeling the typical dinner price at a restaurant, and we know the quality of the food, and also how good the service is.

y = price $\quad x_1$ = food quality (score from 0 to 30) $\quad x_2$ = service quality (0-30)

We could fit two different simple linear regression models, one using each predictor.

could fit 2 models: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 * \text{food}$ $\quad \hat{y} = \hat{\beta}'_0 + \hat{\beta}'_1 * \text{service}$

(different coefficients $\hat{\beta}_0, \hat{\beta}'_0$ & $\hat{\beta}_1, \hat{\beta}'_1$)

But this isn't entirely satisfactory. It would give us an estimate of the relationship between (price and food quality) and (price and service quality).

$\hat{\beta}_1$ estimates relationship btwn price & food quality

$\hat{\beta}'_1$ estimates " " " & service

But how would we use those models to make predictions?

how to make predictions?

For any particular restaurant, we would get 2 different predictions for the dinner price, which isn't very helpful.

2 models → 2 different $\hat{y}$ predictions (not helpful)

Also, each of the simple linear regression models ignores the information from one of the predictors, so we don't necessarily want to just choose one of the two models to use for prediction.

better approach: combined model, multiple predictors

Instead, we will extend the simple linear regression model so that it can directly accommodate multiple predictors.

Each predictor will have its own slope coefficient in a combined model:

β_0 price $= \beta_0 + \beta_1 *$ food $+ \beta_2 *$ service $+ \varepsilon$

The slope β_1 quantifies the association between food quality and price, and β_2 quantifies the association between service quality & price.

In general, we can have more than 2 predictors:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p + \varepsilon$$

β_k x_k is the kth predictor variable

In some ways the multiple linear regression model is different from what we've already studied, and in some ways it is the same.

data:	price	food	service	data point i:
	y_1	$x_{1,1}$	$x_{1,2}$	(x_{i1}, x_{i2}, y_i)
	y_2	$x_{2,1}$	$x_{2,2}$	

x_{i1}: value of 1st predictor; x_{i2}: value of 2nd predictor

Every observation now has a value for each of the predictors. In the restaurant example, the data point for restaurant i consists of that restaurant's food quality, service quality, and dinner price.

We still assume that there is a true linear relationship in the overall population, and we want to use observed data to estimate the coefficients of the true population model. $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$

The multiple linear regression model also still has a random error term ε associated with each data point.

each observation has random error ε, $E(\varepsilon) = 0$

For any particular value of the predictors, the y variable is a random variable, with expected value equal to the corresponding y value on the (population) regression line:

$$E(Y \mid X_1 = x_1, X_2 = x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

estimating the model: method of least squares

The coefficients are estimated using the same least squares approach that we used for simple linear regression.

We want coefficients that minimize the sum of squared residuals: $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$ minimize:

$$\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2}))^2$$

We can't really visualize this in a helpful way anymore, but the core idea is the same: we want a line that is "closest" to the observed data points, and "closest" means that it minimizes the sum of squared residuals.

On the topic of residuals - we can also calculate R^2 for multiple linear regression models.

still true: $R^2 = 1 - \dfrac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$ % explained variation in y

And this quantity still corresponds to the percentage of variation in observed y values that is explained by the linear model.

Next, we will discuss something that is different for the multiple linear regression model: the interpretation of the model coefficients.

Slope interpretation is different for multiple regression.

Suppose we fit a regression model to the restaurant data and obtained these estimates:

$$\widehat{price} = -21 + 1.50\,food + 1.70\,service$$

When interpreting the coefficients in this fitted model, we must be mindful of the multiple predictors involved.

The estimated slope of 1.50 for food quality tells us that we think restaurants with a 1-point difference in food score, and the same service score, have a dinner price that differs by $1.50, on average

$\hat{\beta}_1 = 1.50$: 1 pt diff in food score, $1.50 diff in dinner price, on avg, for restaurants w/SAME service score

Likewise, we estimate that restaurants with equal food quality but a 1-point difference in service quality have dinner prices that differ by $1.70, on average.

$\hat{\beta}_2 = 1.70$: same food score but 1pt diff in service score, & avg $1.70 diff in dinner price

In general, the coefficient β_k is the average difference in response y between observations with a one-unit difference in the predictor x_k, but equal values for all the other predictors. Sometimes this is phrased as "with all other predictors held constant."

Generally: β_k = avg diff in y btwn observations w/ 1-unit diff in predictor x_k, but same values of other predictors

Depending on the real-world context of the problem, it might not make sense for data points to have different values of the k-th predictor but same value for the rest. Don't worry about that for now.

Using multiple predictors in a single model can change our estimate of the relationship between a certain predictor and the response, compared to a simple linear regression using only that one predictor. This is not just a feature of how we estimate the model, either. Incorporating new information can change things fundamentally. (& logically)

estimate:

$\hat{price} = -17.80 + 2.94 * food$ vs $\hat{price} = -21 + 1.50\,food + 1.70\,service$

If we fit a linear regression model to the restaurant data using only food quality as a predictor, the estimated coefficient for food quality is almost twice as large. If we only had access to the food quality info, we would give it more weight when estimating price.

est. slope for food quality changes when service quality is included

In general, the estimated slope coefficient will be different, depending on what other predictor variables are included in the model.

generally: coefficient $\hat{\beta}_k$ for predictor x_k differs depending on what ELSE is included in model

We can consider a few oversimplified examples to illustrate this idea.

① fitting a regression to estimate the # of shark attacks in a particular beach community over time, based on ice cream sales.

ex] y = # shark attacks
x_1 = ice cream sales ($)

$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1$

$\hat{\beta}_1$ will be positive

This would show a positive association.
But if we also included outdoor temperature as a predictor, the coefficient for ice cream sales in the new model will be close to zero (because the observed correlation between shark attacks and ice cream sales was mainly explained by temperature). When we include an additional predictor, the new predictor may be more useful for predicting the response.

x_2 = temperature

$\hat{y} = \hat{\beta}_0' + \hat{\beta}_1' x_1 + \hat{\beta}_2 x_2$

$\hat{\beta}_1'$ close to 0; $\hat{\beta}_2$ positive

② Another example - predicting the price of a house, in a certain region, using the # of bedrooms

y = house price (1000s)
x_1 = # bedrooms

$\hat{y} = 220 + 30 x_1$

more bedrooms — more expensive

Looking at that model, we would see that houses with more bedrooms tend to be more expensive.

But if we also include square footage of the house as a predictor...

X_2 = house size (sq ft)

$\hat{y} = 10 - 5x_1 + 0.175x_2$

for same SIZE houses, more bedrooms LESS expensive on avg

In the new model, the estimated coefficient for # bedrooms is negative, instead of positive. Among houses w/same square footage, the houses with more bedrooms tend to be less expensive. The idea here is that all else equal, having 6 very small bedrooms is less desirable than 3-4 larger bedrooms.

~ computational example in notebook ~